Proof extraction from a failed LTL tableau

a uniform loop-invariant construction

Abstract

Ben-Ari proves completeness of his Hilbert system $\mathcal{L}$ for propositional temporal logic (Theorem 12.18) by transforming a failed semantic tableau into an $\mathcal{L}$-derivation, but carries out the essential case — a strongly connected component that is not self-fulfilling — only on the single example $\Box p \to \bigcirc \Box p$. This note gives the construction uniformly. The loop invariant is taken to be the disjunction of the characteristic formulas of the states *reachable* from the state being refuted, rather than of a strongly connected component; this makes the argument independent of any decomposition into components, and makes the induction follow the elimination order of the search. Everything below is finitary: no infinite path is ever constructed, and no appeal is made to the principle that an infinite path in a finite graph eventually remains within one component. The result is a total function returning either a derivation of φ or a finite countermodel.

Introduction

The point of the construction below is not only that Ben-Ari's classical step is replaced by a finite pigeonhole. It is that the elimination procedure computes both certificates at once: an eliminated state v yields a derivation of $\neg\chi_v$, a surviving state yields a lasso satisfying χ_v, and the procedure produces whichever applies as it runs. The revealing case is (R2), where the failure of an eventuality *automatically* produces an inductive invariant, namely the disjunction B of the characteristic formulas of the states still reachable from v, satisfying $\vdash_{\mathcal{L}} B \to \bigcirc B$; the axiom (Ind) then turns this finite invariant into a proof of $\Box A_j$, which refutes v. So the decision algorithm already contains the Hilbert-style proof implicitly, and extraction is a matter of reading it off rather than reconstructing it.

1 The system $\mathcal{L}$

Formulas are built from propositional variables by $\neg$, $\to$, $\bigcirc$ and $\Box$; we write $\Diamond A$ for $\neg\Box\neg A$, and use $\wedge, \vee, \leftrightarrow$ as the usual abbreviations.

The axioms of $\mathcal{L}$ are

(P1)	$A \to (B \to A)$
(P2)	$\big(A \to (B \to C)\big) \to \big((A \to B) \to (A \to C)\big)$
(P3)	$(\neg B \to \neg A) \to (A \to B)$
(K$\Box$)	$\Box(A \to B) \to (\Box A \to \Box B)$
(K$\bigcirc$)	$\bigcirc(A \to B) \to (\bigcirc A \to \bigcirc B)$
(Exp)	$\Box A \to (A \wedge \bigcirc A \wedge \bigcirc \Box A)$
(Ind)	$\Box(A \to \bigcirc A) \to (A \to \Box A)$
(Lin)	$\bigcirc \neg A \leftrightarrow \neg \bigcirc A$

with modus ponens and generalisation ($\vdash_{\mathcal{L}} A$ implies $\vdash_{\mathcal{L}} \Box A$).

Ben-Ari takes as propositional axiom the scheme (PC), "every substitution instance of a propositional tautology". We replace it by (P1)–(P3). As an axiom, (PC) carries a side condition that must be formalised — a tautology checker and a substitution relation — before the notion of derivation can be defined at all, and it then obliges one to relate evaluation to substitution in the soundness proof. See Remark 6.3.

No completeness theorem for propositional logic is needed to replace it. The propositional reasoning used below is not arbitrary: it consists of the following fixed schemes, each with a short direct derivation from (P1)–(P3).

We fix a propositional variable P_0 and put $\bot := \neg(P_0 \to P_0)$.

Lemma 1.1 (Propositional infrastructure)**.**

(L0) $\vdash_{\mathcal{L}} \neg\bot$, *and* $\vdash_{\mathcal{L}} \bot \to C$ *for every* C.

(L1) $\vdash_{\mathcal{L}} A \to A$; *if* $\vdash_{\mathcal{L}} A \to B$ *and* $\vdash_{\mathcal{L}} B \to C$ *then* $\vdash_{\mathcal{L}} A \to C$; *if* $\vdash_{\mathcal{L}} B$ *then* $\vdash_{\mathcal{L}} A \to B$.

(L2) $\vdash_{\mathcal{L}} (A_1 \wedge \cdots \wedge A_n) \to A_i$; *and if* $\vdash_{\mathcal{L}} C \to A_i$ *for all* i *then* $\vdash_{\mathcal{L}} C \to (A_1 \wedge \cdots \wedge A_n)$.

(L3) $\vdash_{\mathcal{L}} A \to (\neg A \to B)$.

(L4) $\vdash_{\mathcal{L}} A \vee \neg A$.

(L5) $\vdash_{\mathcal{L}} A_i \to (A_1 \vee \cdots \vee A_n)$; *and if* $\vdash_{\mathcal{L}} A_i \to C$ *for every* i *then* $\vdash_{\mathcal{L}} (A_1 \vee \cdots \vee A_n) \to C$.

(L6) If $\vdash_{\mathcal{L}} \bigvee_i C_i$ *then* $\vdash_{\mathcal{L}} \bigvee_i (C_i \wedge A) \vee \bigvee_i (C_i \wedge \neg A)$.

(L7) $\vdash_{\mathcal{L}} \neg A \to (A \to B)$ *and* $\vdash_{\mathcal{L}} B \to (A \to B)$.

Proof. (L0) is $\vdash_{\mathcal{L}} \neg\neg(P_0 \to P_0)$ from (L1) and (P3), with the second part by (L3). (L1) is the standard consequence of (P1),(P2); (L3),(L4),(L7) follow from (P1)–(P3). (L2) and (L5) are proved by induction on n using (L1). (L6) follows from (L4) and (L5) by induction on the list of disjuncts. □

Only (L2) and (L5) are indexed by a list; the rest are of fixed size. We read $\bigvee_{i \in I} A_i$ as $\bot$ when $I = \varnothing$, so that the two clauses of (L5) at $n = 0$ are exactly the two clauses of (L0); every disjunction below is to be understood with this convention, and the empty case genuinely occurs (see Case (R1) of Theorem 4.1). All uses occur under a common antecedent, so they compose by lifting under that antecedent; no deduction theorem is required.

Lemma 1.2 (Derived rules)**.**

(D1) If $\vdash_{\mathcal{L}} A$ *then* $\vdash_{\mathcal{L}} \bigcirc A$.

(D2) If $\vdash_{\mathcal{L}} A \to B$ *then* $\vdash_{\mathcal{L}} \bigcirc A \to \bigcirc B$.

(D3) If $\vdash_{\mathcal{L}} A \to B$ *then* $\vdash_{\mathcal{L}} \Box A \to \Box B$.

(D4) $\vdash_{\mathcal{L}} \bigcirc(A \vee B) \leftrightarrow \bigcirc A \vee \bigcirc B$, *and likewise for* $\wedge$.

(D5) If $\vdash_{\mathcal{L}} A \to \bigcirc A$ *then* $\vdash_{\mathcal{L}} A \to \Box A$.

(D6) $\vdash_{\mathcal{L}} A \wedge \bigcirc \Box A \to \Box A$.

Proof. (D1) From $\vdash_{\mathcal{L}} A$, generalisation gives $\vdash_{\mathcal{L}} \Box A$; (Exp) with (L2) gives $\vdash_{\mathcal{L}} \Box A \to \bigcirc A$; modus ponens.

(D2) From $\vdash_{\mathcal{L}} A \to B$, (D1) gives $\vdash_{\mathcal{L}} \bigcirc(A \to B)$, then (K$\bigcirc$).

(D3) Same with generalisation and (K$\Box$).

(D4) Read $A \vee B$ as $\neg A \to B$. By (K$\bigcirc$), $\vdash_{\mathcal{L}} \bigcirc(\neg A \to B) \to (\bigcirc \neg A \to \bigcirc B)$; by (Lin) $\bigcirc \neg A$ may be replaced by $\neg \bigcirc A$, giving $\vdash_{\mathcal{L}} \bigcirc(A \vee B) \to \bigcirc A \vee \bigcirc B$. Conversely $\vdash_{\mathcal{L}} A \to A \vee B$ gives $\vdash_{\mathcal{L}} \bigcirc A \to \bigcirc(A \vee B)$ by (D2), and symmetrically for B; conclude by (L5). The case of $\wedge$ is dual.

(D5) From $\vdash_{\mathcal{L}} A \to \bigcirc A$, generalisation gives $\vdash_{\mathcal{L}} \Box(A \to \bigcirc A)$, and (Ind) gives $\vdash_{\mathcal{L}} A \to \Box A$.

(D6) Put $C := A \wedge \bigcirc \Box A$. From $\vdash_{\mathcal{L}} \Box A \to A$ (Exp, L2) and (D2), $\vdash_{\mathcal{L}} \bigcirc \Box A \to \bigcirc A$; from $\vdash_{\mathcal{L}} \Box A \to \bigcirc \Box A$ (Exp, L2) and (D2), $\vdash_{\mathcal{L}} \bigcirc \Box A \to \bigcirc \bigcirc \Box A$. Hence $\vdash_{\mathcal{L}} C \to \bigcirc A \wedge \bigcirc \bigcirc \Box A$, which by (D4) is $\vdash_{\mathcal{L}} C \to \bigcirc C$. By (D5), $\vdash_{\mathcal{L}} C \to \Box C$; since $\vdash_{\mathcal{L}} C \to A$, (D3) gives $\vdash_{\mathcal{L}} \Box C \to \Box A$. □

2 States and transitions

Fix a formula φ and let $\mathrm{Sub}(\varphi)$ be its set of subformulas, a finite set computed by structural recursion.

Remark 2.1. No closure operation is needed. One might expect to work over a Fischer–Ladner closure — $\mathrm{Sub}(\varphi)$ enlarged so that $\Box A$ brings $\bigcirc \Box A$ with it — so that the unfolding $\Box A \leftrightarrow A \wedge \bigcirc \Box A$ becomes a local condition on a single state. It is cleaner to put the unfolding into the *transition relation* instead, as in the automata-theoretic presentation: the auxiliary formulas $\bigcirc \Box A$ then never enter the state space, Sub replaces a fixpoint computation by a structural recursion, and the state space $2^{|\mathrm{Sub}(\varphi)|}$ is smaller. Persistence (Lemma 2.8) becomes true by definition rather than by proof.

Definition 2.2. A *valuation* is a function $v : \mathrm{Sub}(\varphi) \to \{0, 1\}$; write $A \in v$ for $v(A) = 1$ and $A \notin v$ for $v(A) = 0$. A valuation is a *state* if

(S1) it respects the connectives: for $\neg A \in \mathrm{Sub}(\varphi)$, $v(\neg A) = 1 - v(A)$; for $A \to B \in \mathrm{Sub}(\varphi)$, $v(A \to B) = 1$ iff $v(A) = 0$ or $v(B) = 1$;

(S2) for $\Box A \in \mathrm{Sub}(\varphi)$: if $\Box A \in v$ then $A \in v$.

$\mathrm{St}(\varphi)$ denotes the finite, decidable set of states.

Definition 2.3. For states v, w put $v \to w$ iff

(T1) for every $\bigcirc B \in \mathrm{Sub}(\varphi)$: $\bigcirc B \in v$ iff $B \in w$;

(T2) for every $\Box A \in \mathrm{Sub}(\varphi)$: if $\Box A \in v$ then $\Box A \in w$;

(T3) for every $\Box A \in \mathrm{Sub}(\varphi)$: if $\Box A \notin v$ and $A \in v$ then $\Box A \notin w$.

(T2) and (T3) are the two halves of $\Box A \leftrightarrow A \wedge \bigcirc \Box A$ with $\bigcirc \Box A$ read as "$\Box A$ at the successor"; (S2) is the remaining half.

Definition 2.4. The *characteristic formula* of a valuation v is

$$\chi_v := \bigwedge_{A \in \mathrm{Sub}(\varphi)} A^{v(A)}, \qquad A^1 := A,\ A^0 := \neg A,$$

and for $U \subseteq \mathrm{St}(\varphi)$, $\beta_U := \bigvee_{v \in U} \chi_v$ with $\beta_\emptyset := \bot$.

Note that when $\Box A \notin v$ the literal $\neg \Box A$ occurs in χ_v, although $\neg \Box A$ need not belong to $\mathrm{Sub}(\varphi)$. We shall always speak of $\Box A \notin v$, never of $\neg \Box A \in v$.

Lemma 2.5 (Non-states are refutable)**.** *If a valuation v is not a state then $\vdash_{\mathcal{L}} \neg \chi_v$.*

Proof. Each violation is a contradiction among finitely many conjuncts of χ_v, extracted by (L2) and closed by (L3); we list them.

If (S1) fails at $\neg A$: either $\neg A, A$ are both conjuncts of χ_v, or $\neg\neg A, \neg A$ both are.

If (S1) fails at $A \to B$ with $v(A \to B) = 1$: then $A \to B$, A, $\neg B$ are conjuncts, and modus ponens under the antecedent gives both B and $\neg B$. If it fails with $v(A \to B) = 0$: then $\neg(A \to B)$ is a conjunct together with $\neg A$ or with B, and (L7) gives $A \to B$.

If (S2) fails: $\Box A$ and $\neg A$ are conjuncts, and (Exp) with (L2) gives $\vdash_{\mathcal{L}} \Box A \to A$. □

Lemma 2.6 (State completeness)**.** $\vdash_{\mathcal{L}} \beta_{\mathrm{St}(\varphi)}$.

Proof. Enumerate $\mathrm{Sub}(\varphi) = \{A_1, \ldots, A_n\}$ and let D_k be the disjunction of $\bigwedge_{i \le k} A_i^{u(A_i)}$ over all $u : \{A_1, \ldots, A_k\} \to \{0, 1\}$. Then $\vdash_{\mathcal{L}} D_0$, and $\vdash_{\mathcal{L}} D_k$ implies $\vdash_{\mathcal{L}} D_{k+1}$ by (L6) with $A := A_{k+1}$. Hence $\vdash_{\mathcal{L}} D_n = \bigvee_v \chi_v$, ranging over *all* valuations. By Lemma 2.5 the disjuncts belonging to non-states are refutable, so they are dropped by (L3) and (L5). □

The derivation is thus built by structural induction along $\mathrm{Sub}(\varphi)$; no fresh variables and no substitution occur, in contrast with the usual formulation via (PC).

Lemma 2.7 (Step)**.** *For every state v, $\vdash_{\mathcal{L}} \chi_v \to \bigcirc\big(\beta_{\{w : v \to w\}}\big)$.*

Proof. By Lemma 2.6 and (D1), $\vdash_{\mathcal{L}} \bigcirc \beta_{\mathrm{St}(\varphi)}$, hence $\vdash_{\mathcal{L}} \bigvee_{w \in \mathrm{St}(\varphi)} \bigcirc \chi_w$ by (D4). Let w be a state with $v \not\to w$; we show $\vdash_{\mathcal{L}} \neg(\chi_v \wedge \bigcirc \chi_w)$.

If (T1) fails there is $\bigcirc B \in \mathrm{Sub}(\varphi)$ with $\bigcirc B \in v$, $B \notin w$ or with $\bigcirc B \notin v$, $B \in w$. In the first case $\vdash_{\mathcal{L}} \chi_v \to \bigcirc B$ and $\vdash_{\mathcal{L}} \chi_w \to \neg B$, so $\vdash_{\mathcal{L}} \bigcirc \chi_w \to \bigcirc \neg B$ by (D2) and $\vdash_{\mathcal{L}} \bigcirc \neg B \to \neg \bigcirc B$ by (Lin); the second case is symmetric.

If (T2) fails then $\Box A \in v$ and $\Box A \notin w$. Then $\vdash_{\mathcal{L}} \chi_v \to \Box A$, and (Exp) gives $\vdash_{\mathcal{L}} \chi_v \to \bigcirc \Box A$; while $\vdash_{\mathcal{L}} \chi_w \to \neg \Box A$ gives $\vdash_{\mathcal{L}} \bigcirc \chi_w \to \neg \bigcirc \Box A$ by (D2) and (Lin).

If (T3) fails then $\Box A \notin v$, $A \in v$ and $\Box A \in w$. By (L2), $\vdash_{\mathcal{L}} \chi_v \to A$ and $\vdash_{\mathcal{L}} \chi_v \to \neg \Box A$, and $\vdash_{\mathcal{L}} \chi_w \to \Box A$ gives $\vdash_{\mathcal{L}} \bigcirc \chi_w \to \bigcirc \Box A$ by (D2). Hence

$$\vdash_{\mathcal{L}} (\chi_v \wedge \bigcirc \chi_w) \to (A \wedge \bigcirc \Box A),$$

and (D6) turns this into $\vdash_{\mathcal{L}} (\chi_v \wedge \bigcirc \chi_w) \to \Box A$, while $\vdash_{\mathcal{L}} (\chi_v \wedge \bigcirc \chi_w) \to \neg \Box A$; conclude by (L3).

This last case is the proof-theoretic reason for including (T3) explicitly in Definition 2.3: it is precisely the condition whose violation (D6) refutes.

Dropping these disjuncts by (L3),(L5) yields $\vdash_{\mathcal{L}} \chi_v \to \bigvee_{v \to w} \bigcirc \chi_w$, which is the claim by (D4). □

Lemma 2.8 (Persistence)**.** *If $v \to w$, $\Box A \notin v$ and $A \in v$, then $\Box A \notin w$.*

Proof. This is (T3). □

3 The elimination procedure

Let $\Box A_1, \ldots, \Box A_m$ enumerate the members of $\mathrm{Sub}(\varphi)$ of the form $\Box A$. Say that the j-th eventuality is *pending* at a state v if $\Box A_j \notin v$, and *fulfilled* at w if $A_j \notin w$.

For $U \subseteq \mathrm{St}(\varphi)$ and $v \in U$, let $\mathrm{Reach}_U(v)$ be the set of states reachable from v by $\to$-steps inside U, including v.

Definition 3.1 (Elimination)**.** Set $U_0 := \mathrm{St}(\varphi)$. Given U_k, remove from it every state v to which one of the following applies, and call the result U_{k+1}:

(R1) v has no $\to$-successor in U_k;

(R2) for some j, the j-th eventuality is pending at v and is fulfilled at no state of $\mathrm{Reach}_{U_k}(v)$.

Stop when $U_{k+1} = U_k$; write U_∞ for the result and $R := \mathrm{St}(\varphi) \setminus U_\infty$.

Both conditions are decidable, each round removes at least one state unless it is the last, and $\mathrm{St}(\varphi)$ is finite, so the procedure terminates in at most $|\mathrm{St}(\varphi)|$ rounds.

Remark 3.2. Condition (R2) may be read as a greatest fixed point, and it is worth doing so. Put $W_j^0 := \{w \in U_k : A_j \in w\}$, and let W_j be the largest subset of W_j^0 closed under $\to$-successors lying in U_k — computed by deleting from W_j^0 every state having a successor in U_k outside it, and repeating. Then, for j pending at v,

$$A_j \in w \ \text{ for every } w \in \mathrm{Reach}_{U_k}(v) \qquad \Longleftrightarrow \qquad v \in W_j,$$

since $\mathrm{Reach}_{U_k}(v)$ is the least closed set containing v and W_j the largest closed subset of W_j^0.

The reformulation costs nothing and buys step (ii) of Theorem 4.1. The only properties of W used there are that A_j holds throughout it, that $v \in W$, and that it is closed under $\to$-successors lying in U_k; taking $W := W_j$ the first is inclusion in W_j^0, the second is (R2) itself, and the third holds *by construction*, where for $\mathrm{Reach}_{U_k}(v)$ it holds only once one knows that the iteration computing that set has converged. The proof of the theorem below goes through verbatim with either reading.

4 The extraction theorem

This is the content that Ben-Ari gives only by example.

Theorem 4.1 (Extraction)**.** $\vdash_{\mathcal{L}} \neg\chi_v$ *for every* $v \in R$.

Proof. By induction on the round k at which v is removed; the induction hypothesis is that $\vdash_{\mathcal{L}} \neg\chi_u$ for every $u \in \mathrm{St}(\varphi) \setminus U_k$.

Case (R1). By Lemma 2.7, $\vdash_{\mathcal{L}} \chi_v \to \bigcirc \beta_S$ with $S = \{w : v \to w\}$. Every $w \in S$ lies outside U_k, since by (R1) none lies in U_k; so $\vdash_{\mathcal{L}} \neg\chi_w$ by hypothesis for each, whence $\vdash_{\mathcal{L}} \neg\beta_S$ by (L5). Here S may be empty — indeed it is whenever v has no successors at all — and then $\beta_S = \bot$ and $\vdash_{\mathcal{L}} \neg\beta_S$ is (L0); the two cases are uniform under the convention fixed after Lemma 1.1. By (D1) $\vdash_{\mathcal{L}} \bigcirc \neg\beta_S$, and by (Lin) $\vdash_{\mathcal{L}} \neg \bigcirc \beta_S$. Therefore $\vdash_{\mathcal{L}} \neg\chi_v$ by (L3).

Case (R2). Fix j as in (R2) and put

$$W := \mathrm{Reach}_{U_k}(v), \qquad B := \beta_W.$$

B is the loop invariant.

(i) $\vdash_{\mathcal{L}} B \to A_j$. By the hypothesis of (R2) the j-th eventuality is fulfilled at no state of W, that is $A_j \in w$ for every $w \in W$. So A_j is a conjunct of χ_w and $\vdash_{\mathcal{L}} \chi_w \to A_j$ by (L2); conclude by (L5).

(ii) $\vdash_{\mathcal{L}} B \to \bigcirc B$. Let $w \in W$ and $S_w := \{w' : w \to w'\}$. By Lemma 2.7, $\vdash_{\mathcal{L}} \chi_w \to \bigcirc \beta_{S_w}$. For $w' \in S_w \setminus U_k$ we have $\vdash_{\mathcal{L}} \neg\chi_{w'}$ by hypothesis; for $w' \in S_w \cap U_k$ we have $w' \in W$, since W is closed under $\to$-successors lying in U_k. Hence $\vdash_{\mathcal{L}} \beta_{S_w} \to B$ by (L3),(L5), this being (L0) when $S_w = \varnothing$, and $\vdash_{\mathcal{L}} \bigcirc \beta_{S_w} \to \bigcirc B$ by (D2). So $\vdash_{\mathcal{L}} \chi_w \to \bigcirc B$ for every $w \in W$; conclude by (L5).

(iii) $\vdash_{\mathcal{L}} B \to \Box A_j$. From (ii) and (D5), $\vdash_{\mathcal{L}} B \to \Box B$; from (i) and (D3), $\vdash_{\mathcal{L}} \Box B \to \Box A_j$.

Now $v \in W$, so $\vdash_{\mathcal{L}} \chi_v \to B$ and hence $\vdash_{\mathcal{L}} \chi_v \to \Box A_j$. Since the eventuality is pending at v, $\neg\Box A_j$ is a conjunct of χ_v, so $\vdash_{\mathcal{L}} \chi_v \to \neg\Box A_j$ by (L2). Therefore $\vdash_{\mathcal{L}} \neg\chi_v$ by (L3). □

Remark 4.2. The invariant is the disjunction over the *reachable* set W, not over a strongly connected component. This is what makes the case uniform: no decomposition into components is needed, edges leaving U_k are absorbed by the induction hypothesis in step (ii), and the pending eventuality is refuted by a single use of (Ind) through (D5).

Ben-Ari's worked example $\varphi = \neg(\Box p \to \bigcirc\Box p)$ is *not* an instance of this case. In the present formulation it is settled by (R1) alone: because $\bigcirc\Box p$ lies in $\mathrm{Sub}(\varphi)$ there, (T1) forces $\Box p \notin w$ at any successor w while (T2) forces $\Box p \in w$, so the states containing φ have no successor at all and are removed in the first round. His tableau needed the loop invariant precisely because it carried $\bigcirc\Box p$ as a member of the closure; here Remark 2.1 has moved that formula into the transition relation, and the contradiction becomes local. A small instance that does reach (R2) is $\varphi = \neg(\Box p \to \Box\bigcirc p)$: the state with $\Box p$ and $\Box\bigcirc p \notin v$ has successors, so (R1) does not apply, while (T2) carries $\Box p$ along every path, (S2) then gives p and (T1) gives $\bigcirc p$ at every state reached, so the eventuality that $\Box\bigcirc p$ leaves pending is fulfilled nowhere.

Remark 4.3. The proof gives more than is stated: the same B refutes every $w \in W$ at which the j-th eventuality is pending, so all of them may be removed in one round, sharing a single derivation.

5 The other branch: a fulfilling lasso

Lemma 5.1. *Every $v \in U_\infty$ has a $\to$-successor in U_∞, and for every $v \in U_\infty$ and every j pending at v, some state of* $\mathrm{Reach}_{U_\infty}(v)$ *fulfils the j-th eventuality.*

Proof. Otherwise (R1) or (R2) would apply and the procedure would not have stopped. □

Definition 5.2. A *round* from $u \in U_\infty$ is the finite path in U_∞ obtained as follows. Let $u_0 := u$. For $j = 1, \ldots, m$: if the j-th eventuality is pending at u_{j-1}, let u_j be the endpoint of a path in U_∞ from u_{j-1} to a state fulfilling it, which exists by Lemma 5.1; otherwise $u_j := u_{j-1}$ by the empty path. Finally let u_{m+1} be any $\to$-successor of u_m in U_∞, which exists by the first half of Lemma 5.1. The round ends at u_{m+1}, and traverses at least one transition.

The last step is not decoration. Without it a round is empty whenever no eventuality is pending at u, and that case occurs: for $\varphi = \neg(\Box p \to \Box \bigcirc p)$ the surviving state at which p holds forever leaves nothing pending. Then all the boundaries $b_0, b_1, \ldots$ of Proposition 5.3 coincide without a single transition being traversed, the "cycle" obtained there has length zero, and the unwinding after it divides by $q - p = 0$. With the last step every round traverses at least one transition, so a repetition among the boundaries yields a cycle of length at least one. Nothing else in the argument is affected: the fulfilment property proved below concerns the states a round visits, and adding a state to the end only enlarges that set.

Proposition 5.3. *If $U_\infty \neq \emptyset$ then from every $v_0 \in U_\infty$ there is a* fulfilling lasso*: a finite path $v_0 \to \cdots \to v_p$ followed by a cycle $v_p \to \cdots \to v_q = v_p$ such that every eventuality pending at a state of the cycle is fulfilled at a state of the cycle.*

Proof. Concatenate rounds from v_0, giving boundaries $b_0 = v_0, b_1, b_2, \ldots$. Among $b_0, \ldots, b_{|U_\infty|}$ two coincide, say $b_a = b_c$ with $a < c$; this is the pigeonhole principle for an explicitly given finite list, so the indices are found by inspection. Take the prefix up to b_a and the cycle from b_a to b_c, which by Definition 5.2 traverses at least $c - a$ transitions and so is a genuine cycle.

It is worth saying how the inspection goes, because it needs no pigeonhole *lemma*. Generate the boundaries one at a time, carrying alongside them the set S of states of U_∞ not yet used as a boundary, so that every state of U_∞ is either an earlier boundary or in S. A newly generated boundary is either one already seen — and then the cycle closes — or a fresh one, and then it leaves S, which can happen at most $|U_\infty|$ times. The pigeonhole is thus the invariant of a terminating search rather than a separate principle to be invoked.

Let the j-th eventuality be pending at a state s of the cycle. Travelling along the cycle from s, either it is fulfilled at some state — and we are done — or by Lemma 2.8 it is still pending at the endpoint $b_c = b_a$, hence at the start of round $a{+}1$, which fulfils it by construction. Either way it is fulfilled inside the cycle. □

The lasso is unwound into the infinite sequence $v_0, v_1, v_2, \ldots$ with $v_i = v_{p+((i-p) \bmod (q-p))}$ for $i \geq p$.

Lemma 5.4 (Global fulfillment)**.** *For every i and every $\Box A \in \mathrm{Sub}(\varphi)$ with $\Box A \notin v_i$, there is $j \geq i$ with $A \notin v_j$.*

Proof. Suppose first that $i \geq p$, so v_i lies on the cycle. By Proposition 5.3 the eventuality is fulfilled at some state of the cycle, and since the cycle repeats, that state occurs at some index $j \geq i$.

Now suppose $i < p$. If $A \notin v_k$ for some k with $i \leq k < p$ we are done. Otherwise $A \in v_k$ for all such k, and Lemma 2.8 gives $\Box A \notin v_{k+1}$ successively, so $\Box A \notin v_p$; apply the previous case at $p \geq i$. □

The lasso is a finite structure with a total successor function, so $\Box$ and $\Diamond$ evaluate on it by conjunction and disjunction over the finite forward orbit of a state; write $[\![\cdot]\!]_s$ for that Boolean evaluation. One checks by induction on derivations that $\vdash_{\mathcal{L}} C$ implies $[\![C]\!]_s = 1$ at every state of every lasso — soundness — and by induction on formulas, using (T1) for $\bigcirc$, (S2) and (T2) for $\Box A \in v_i$, and Lemma 5.4 for $\Box A \notin v_i$, that

$$A \in v_i \iff [\![A]\!]_{v_i} = 1 \qquad (A \in \mathrm{Sub}(\varphi)),$$

which is Hintikka's lemma.

6 Decidability

Theorem 6.1. *There is a total computable function taking φ to an element of*

$$\mathrm{Deriv}_{\mathcal{L}}(\varphi) \;+\; \mathrm{LassoCountermodel}(\varphi).$$

Proof. Run the elimination of Definition 3.1 on $\mathrm{Sub}(\neg\varphi)$ and ask whether some $v_0 \in U_\infty$ has $\neg\varphi \in v_0$. This is decidable, U_∞ being finite.

If there is one, Proposition 5.3 gives a fulfilling lasso through v_0, and Hintikka's lemma gives $[\![\neg\varphi]\!]_{v_0} = 1$, so $[\![\varphi]\!]_{v_0} = 0$. By soundness $\vdash_{\mathcal{L}} \varphi$ would give $[\![\varphi]\!]_{v_0} = 1$; the lasso is therefore a countermodel.

If there is none, every state containing $\neg\varphi$ lies in R, so $\vdash_{\mathcal{L}} \neg\chi_v$ for each by Theorem 4.1. Every remaining state contains φ, so $\vdash_{\mathcal{L}} \chi_v \to \varphi$ by (L2). By Lemma 2.6 and (L3),(L5), $\vdash_{\mathcal{L}} \varphi$. □

In particular $\vdash_{\mathcal{L}} \varphi$ is decidable, but the theorem says more: both branches carry a checkable certificate, and neither is obtained by refuting the other.

Remark 6.2. No step above constructs an infinite path, and the principle that an infinite path in a finite graph eventually remains within one strongly connected component — the converse half of Ben-Ari's Theorem 11.44, and the "key insight" of the usual automata-theoretic presentation — is never used. The only combinatorial principle is the pigeonhole principle for an explicitly given finite list, in Proposition 5.3, and even that is not needed as a lemma: organised as in the proof there, it is the termination of a search that removes one state from a finite pool at each round. The two branches of Theorem 6.1 are asymmetric by design: the failure branch returns a derivation rather than an assertion of unsatisfiability, and that is what removes the need for the classical step.

Remark 6.3. On (PC). Read as an axiom scheme, "substitution instance of a propositional tautology" carries a side condition, so a formalisation must define a tautology checker and the substitution relation before it can say what a derivation *is*, and must then prove a substitution lemma relating evaluation to substitution in order to establish soundness. With (P1)–(P3) the derivations form a plain inductive family with ten constructors — the eight axiom schemes, modus ponens and generalisation — and soundness inspects three concrete formulas.

No propositional *completeness* theorem is needed to make up the difference. One might expect to recover (PC) as a derived rule by Kalmár's argument and then discharge each propositional step by a truth-table computation; but Kalmár's argument needs the deduction theorem to discharge its hypotheses, and it is unnecessary here. The propositional facts used are the fixed schemes of Lemma 1.1, of which only (L2) and (L5) are indexed by a list, and the one step that looks as though it needs a tautology checker — state completeness, Lemma 2.6 — is instead a structural induction along $\mathrm{Sub}(\varphi)$.

Remark 6.4 (Formalisation)**.** Everything above has been checked in Agda — `--safe`, `--without-K`, `--exact-split`, no library — in thirty modules and about five thousand lines, with Theorem 6.1 as a total function

$$\mathrm{decide} : (Q : \mathrm{Formula}) \to \mathrm{Deriv}(Q) \;+\; \mathrm{LassoCountermodel}(Q),$$

and with no postulates, no unfilled goals and no termination pragmas. The development runs: $\Box p \to \Box\Box p$ and $\Box(p \to \bigcirc p) \to (p \to \Box p)$ are decided affirmatively, and $\Diamond p \to \Box p$ negatively, returning the lasso on which p recurs without persisting.

Three of the remarks above record places where formalising changed the argument rather than merely transcribing it: the final step in Definition 5.2, without which Proposition 5.3 is false as stated; Remark 3.2, where reading (R2) as a greatest fixed point makes the closure needed in step (ii) of Theorem 4.1 hold by construction; and the reading of the pigeonhole as a search invariant. A fourth is the correction to the remark following Theorem 4.1: Ben-Ari's own example does not exercise case (R2) here, which the machine noticed and the author had not.

Remark 6.5. The same strategy should extend to *until*, but the extension is not purely notational and is not claimed here. One adds $A \mathcal{U} B \leftrightarrow B \vee (A \wedge \bigcirc(A \mathcal{U} B))$ to Definitions 2.2 and 2.3 in the manner of (S2),(T2),(T3), and takes $A \mathcal{U} B$ pending at v when $A \mathcal{U} B \in v$ and $B \notin v$. What has to be checked is that step (i) of Theorem 4.1 still goes through: the obligation is no longer the single formula A_j but the conjunction of A with the persistence of $A \mathcal{U} B$, so the invariant $B = \beta_W$ must be shown to imply both. The analogue of Lemma 2.8 likewise has to carry the side condition that A holds throughout. Half a page of proof, not a remark.